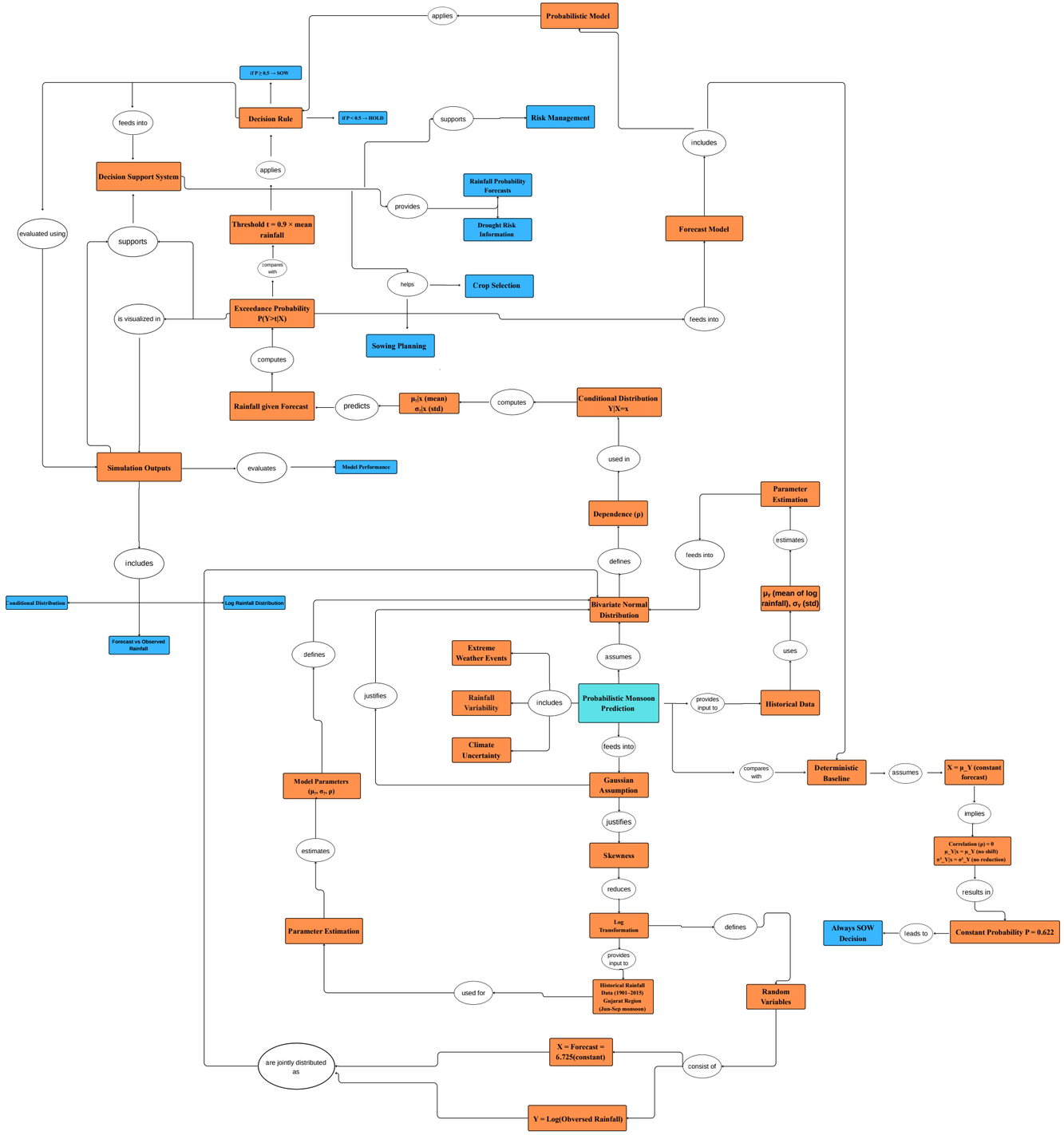
Probabilistic Model
applies
if P ≥ 0.5 → SOW
Decision Rule
if P < 0.5 → HOLD
supports
Risk Management
feeds into
Decision Support System
applies
includes
Rainfall Probability Forecasts
provides
Drought Risk Information
Forecast Model
evaluated using
supports
Threshold t = 0.9 × mean rainfall
compares with
helps
Crop Selection
is visualized in
Exceedance Probability P(Y>t|X)
feeds into
Sowing Planning
computes
Rainfall given Forecast
predicts
μ_Y|X (mean) σ_Y|X (std)
computes
Conditional Distribution Y|X=x
used in
Simulation Outputs
evaluates
Model Performance
Parameter Estimation
Dependence (ρ)
estimates
feeds into
defines
includes
μ_Y (mean of log rainfall), σ_Y (std)
Conditional Distribution
Log Rainfall Distribution
Bivariate Normal Distribution
Forecast vs Observed Rainfall
defines
Extreme Weather Events
assumes
uses
justifies
Rainfall Variability
includes
Probabilistic Monsoon Prediction
provides input to
Historical Data
Climate Uncertainty
feeds into
compares with
Deterministic Baseline
assumes
X = μ_Y (constant forecast)
Model Parameters (μ_Y, σ_Y, ρ)
Gaussian Assumption
implies
justifies
Correlation (ρ) = 0
μ'_Y|x = μ_Y (no shift)
σ'_Y|x = σ'_Y (no reduction)
estimates
Skewness
reduces
results in
Parameter Estimation
Log Transformation
defines
Always SOW Decision
leads to
Constant Probability P = 0.622
provides input to
used for
Historical Rainfall Data (1901–2015) Gujarat Region (Jun-Sep monsoon)
Random Variables
are jointly distributed as
X = Forecast = 6.725(constant)
consist of
Y = Log(Obversed Rainfall)